TERM 2

SCIENCE NOTES

1. WEATHER
2. METALS
3. HOUSE-HOLD CHEMICALS
4. ENERGY
5. HEAT
6. LIGHT
7. SOUND
8. FORCE

@Tz

WEATHER

- ✈ Weather is the condition of the atmosphere over a short period of time.
- ✈ A person who studies weather is called a meteorologist.

Weather changes from day today. It can also change just in hours. For example, in the morning it can be cold and in the afternoon it can be hot.

MEASURING WEATHER CONDITIONS

The elements of weather are;

a) Temperature b) Wind direction c) Wind speed d) Humidity
e) Air pressure f) Cloud cover

1. MEASURING TEMPERATURE

- Temperature is measured with a thermometer.



- It is measured in *degrees Celsius* (°C).
- The thermometer has liquid called mercury inside it. If the temperature is hot, the liquid *expand* and if it is low it *contacts*.

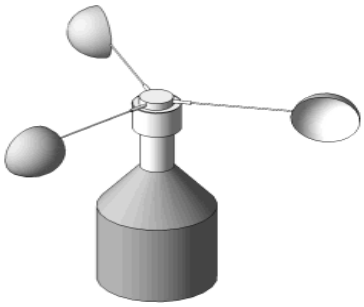
2. MEASURING WIND DIRECTION

- We use a wind vane to show wind direction.

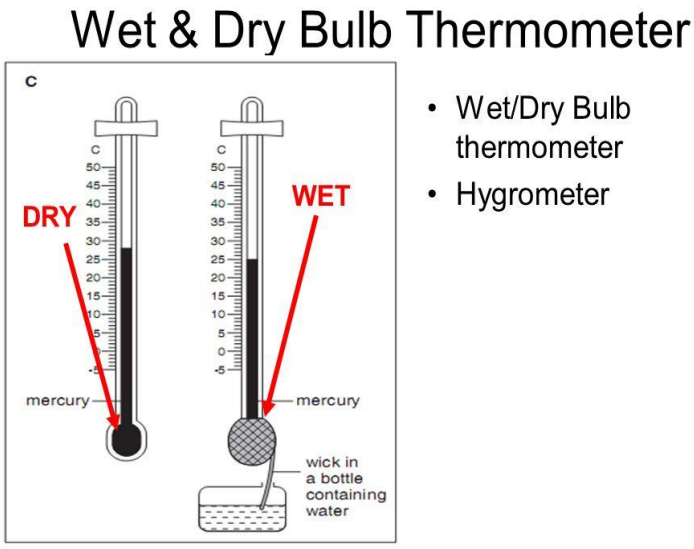


- It is reported in *cardinal directions (north, east, west and south.)* and measured in
- The arrow of the windvane points to the direction from which the wind is blowing from.

3. MEASURING WIND SPEED



- We use an anemometer to measure wind speed *kilometers per hour (km/h).*
- It also measures wind pressure.
- A cup anemometer has a vertical pole with three cups that capture the wind. The number of times the cups spin around per minute is counted electronically

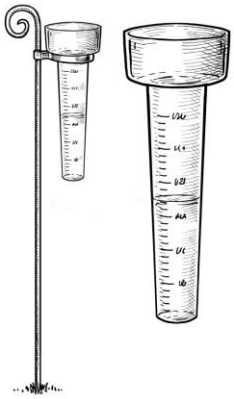


- Wet/Dry Bulb thermometer
- Hygrometer

4. MEASURING HUMIDITY

- Humidity is the quality of *water vapour* in the air.
- We use a *hygrometer* to measure humidity. The simple stone is called a wet and dry bulb hygrometer.
- We read the temperature on both thermometers. The bigger the difference between the readings, the drier the air.

5. MEASURING RAINFALL



- The quality of rain that falls is measured with a rain gauge.
- The rain gauge must be placed in an open space, at least 1m above the ground where there are no obstacles.
- The rainfall is measured in *milliliters (ml).*

6. MEASURING AIR PRESSURE

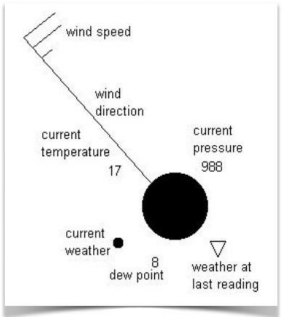


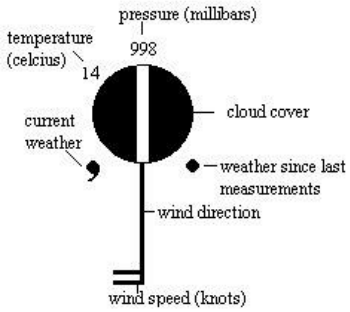
- We use a *barometer* to measure air pressure. It is measured in *millibars (Mb).*
- A high pressure reading shows the air is dry.
- If the air pressure drops, it shows that the air is *moist* and it could indicate rainfall.

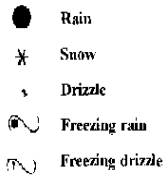
MEASURING WEATHER ELEMENTS

ELEMENTS OF WEATHER	MEASURING INSTRUMENTS	UNIT OF MEASURE
Temperature	Thermometer	Degree Celsius (°C)
Wind Direction	Wind Vane	Cardinal points
Wind Speed	Anemometer	Km/hr
Air Pressure	Barometer	Millibars(Mb)
Cloud Cover	Ceilometers (Observation)	Oktas
Sunshine	Sunshine recorder	Hrs
Rainfall	Rain gauge	Millimeters (mm)

WEATHER SYMBOLS ON A CHART



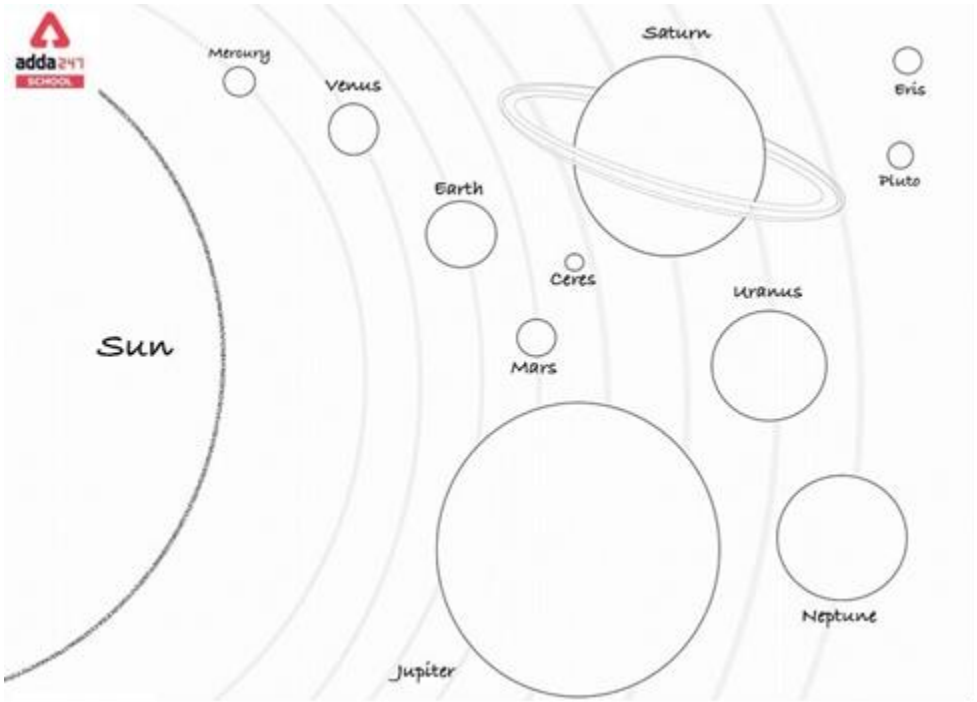






THE SOLAR SYSTEM

a) The Universe



The universe is everything that exists anywhere. The sun, the moon, earth and other planets and starsare also known as the universe. The universe consists of billions of stars.

- A star is a very large ball of very hot gas. Some stars are thousands of times as big as earth and they give of lots of heat and light.
- Planets are small bodies that travel around some stars. They do not produce their own light and can only be seen because the light of a star shines on it.

a. <u>The Solar system</u>

The word solar comes from the word *sol* which means sun. The solar system is made up of the sun and 8 bodies or planets that travel around it.

Why Is Pluto Not Called a Planet Anymore?

In 2003, an astronomer saw a new object beyond Pluto. The astronomer thought he had found a new planet. The object he saw was larger than Pluto. He named the object Eris. Finding Eris caused other astronomers to talk about what makes a planet a "planet."

There is a group of astronomers that names objects in space. This group decided that Pluto was not really a planet because of its size and location in space. So Pluto and objects like it are now called dwarf planets.

- Pluto is also called a *plutoid*. A *plutoid* is a dwarf planet that is farther out in space than the planet Neptune.

Astronomers use *telescopes* to discover new objects like plutoids. Some stars in the sky are brighter while some are very faint.

- The brightest star in the sky at night is often called the *evening star*. This is really the planet Venus and not a star at all.
- Jupiter is the second brightest object in the night sky. Mars can often be seen as a red body. The four inner planets are Mercury, Venus, Earth and Mars. They are all solid and are made of rocky material.
- The next four planets further from the sun are Jupiter, Saturn, Uranus and Neptune. They are made of gas and they are much larger than the inner planets.

THE FOUR INNER PLANETS

i. <u>Mercury</u>

- It is the smallest of the four inner planets and the closest to the sun.
- It is so hot that lead will melt on its surface.
- No life can exist on it (no atmosphere and water).

ii. <u>Venus</u>

- ✓ Venus is about the same size of the Earth.
- ✓ It is very bright in the sky at night and is sometimes called evening star.
- ✓ It is often covered with dense clouds of poisonous gases which keep heat from the sun makingVenus even hotter than Mercury.

iii. <u>Earth</u>

- ➢ The Earth is about 150 million kilometers from the sun.
- ➢ It is not too hot or too cold.
- ➢ There is lots of water on Earth and it has a big layer of air (atmosphere) around it.
- ➢ The Earth takes 365 days to go round the sun once (orbit the sun).
- ➢ The Earth has one moon that orbits around it.
- ➢ It is the only planet capable of supporting life so far.

iv. Mars

- Mars is smaller than Earth and its surface temperature is very low about -23 degrees Celsius.
- The surface is often covered by very thick dusty storms because of strong winds.
- Mars has two moons that orbit around it.

THE FOUR OUTER PLANETS

- They are called *gas giants* because they are mainly made of gases.
- They are very large.
- They are very cold because they are far from the sun.
- They have moons that orbit around them.

i. Jupiter

✓ It is the *largest planet* in the solar system.
✓ It is very far from the sun and it is extremely cold.
✓ It is the second brightest object in the sky after Venus.
✓ It takes 12 years to go right round the sun once. Jupiter has 16 moons.

ii. Saturn

» It has a series of bright rings around it which consist of small pieces of rock.
» Saturn has 18 moons and takes 29 years to go round the sun once.

iii. Uranus

➢ It is about 70 times as large as the Earth.
➢ It has a green\blue colour and 20 moons.
➢ It takes 84 years to orbit the sun.

iv. Neptune

✓ It is the furthest of the gas giants from the sun.
✓ It is blue and has 15 smaller moons.
✓ It takes 165 years to go round the sun once.

THE SUN

➢ It is the nearest star to the Earth.
➢ It is at the centre of the solar system and it is a very large ball of flaming gas.
➢ All the heat and energy onEarth comes from the Sun.

WHY IS EARTH THE ONLY PLANET CAPABLE OF SUPPORTING LIFE

A planet needs three things for organisms to live there, namely:

✓ An *atmosphere* that can be breathed- oxygen and carbon dioxide
✓ A *suitable temperature* – not to hot or too cold
✓ Water

So far the only planet in the solar system that is capable of supporting life is earth because it meets all the requirements.

MATTER AND ENERGY

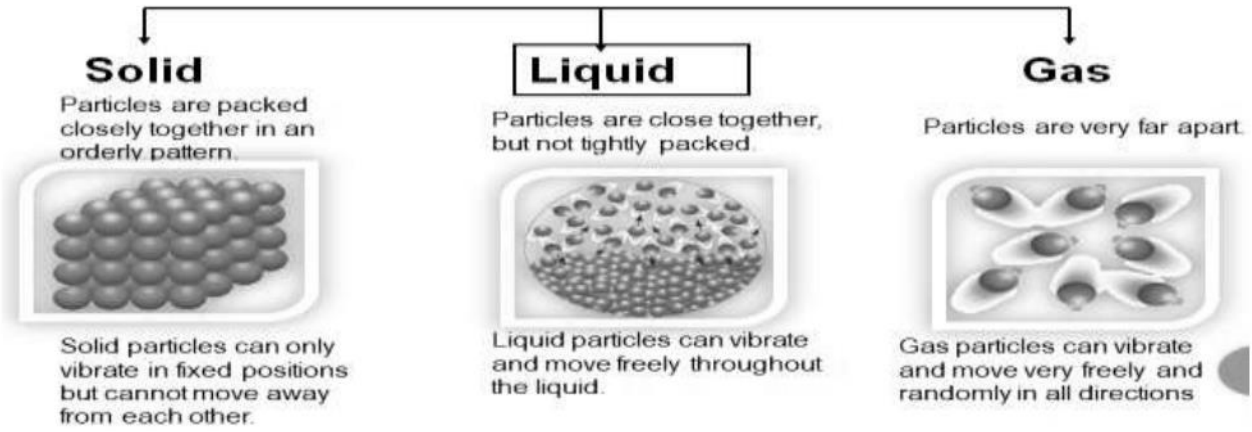
MATTER

What is matter?

It is everything around us, everything we can touch and feel.
Matter is everything that has mass andoccupies space.
Wood, water, iron, air and paper are all matter.

STATES OF MATTER

The states of matter are solid, liquid and gas. The states of matter depend on the arrangement of particles.



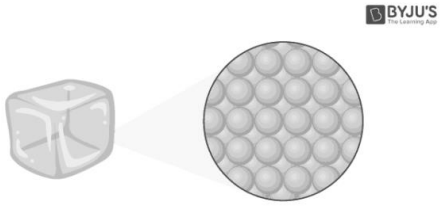
Properties of the different states

- A solid is hard and it keeps its shape. It cannot move around or flow.
- A liquid does not keep its shape; it takes the sharp of the container. It flows from one place to another.
- A gas has no shape or size. It will fill all space in the container.

The properties of matter

✓ Matter is made up of molecules which are very small particles that are too small to see.
✓ These molecules are moving all the time. We say they are vibrating. The hotter they are, the stronger the vibration.
✓ When molecules are very close, they attract one another very strongly and when they are far apart the attraction is very weak.

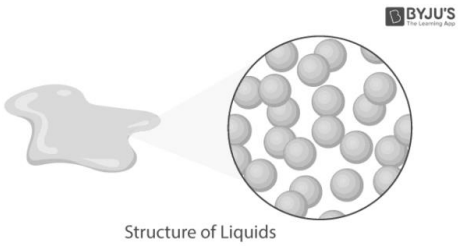
PROPERTIES OF SOLIDS



Structure of solids

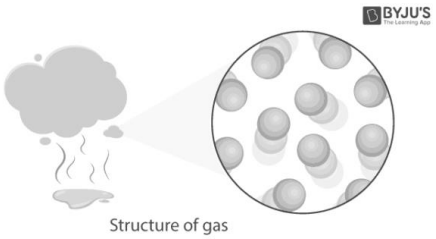
1. The particles in a solid are closely packed together and attract one another very strongly.
2. They are arranged in a regular pattern and have a fixed shape and size.
3. They can only vibrate in their positions.
4. The particles have very little energy.

PROPERTIES OF LIQUIDS



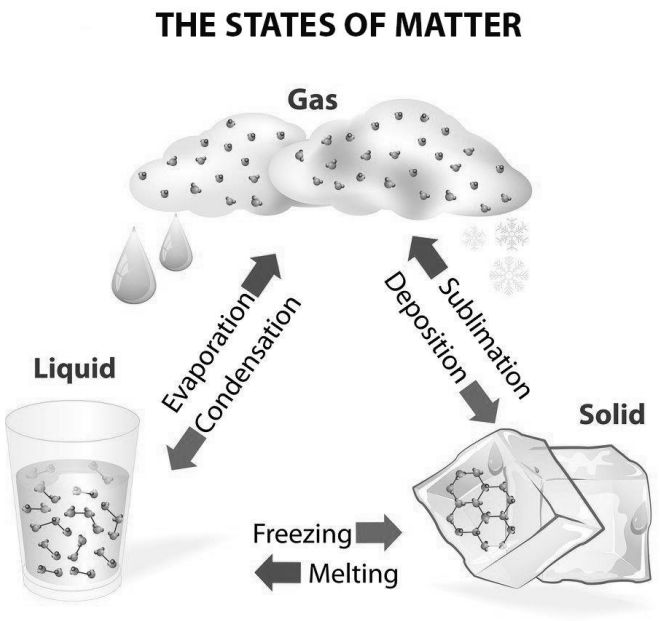
1. The particles are packed close together but still can move.
2. The spaces between the particles in a liquid make it possible for the particles to move and slide over each other.
3. The force of attraction is less strong thus enabling the molecules to move about and take the shape of the container.

PROPERTIES OF GASES



1. The particles in a gas are spaced far apart.
2. The particles have a lot of energy and are able to move around very quickly.
3. The force of attraction between molecules is very weak.
4. Gases have no fixed volume or shape because particles can move around freely.
5. Gases can be squashed together to fit in a small space.

CHANGES IN THE STATES OF MATTER



Matter changes state in various ways:

- Melting - changing from solid to liquid
- Freezing - changing from a liquid to a solid
- Vapourisation - changing from liquid to gas
- Condensation - changing from gas to a liquid
- Deposition - changing from gas to a solid
- Sublimation - changing from a solid to gas

METALS AND NON-METALS

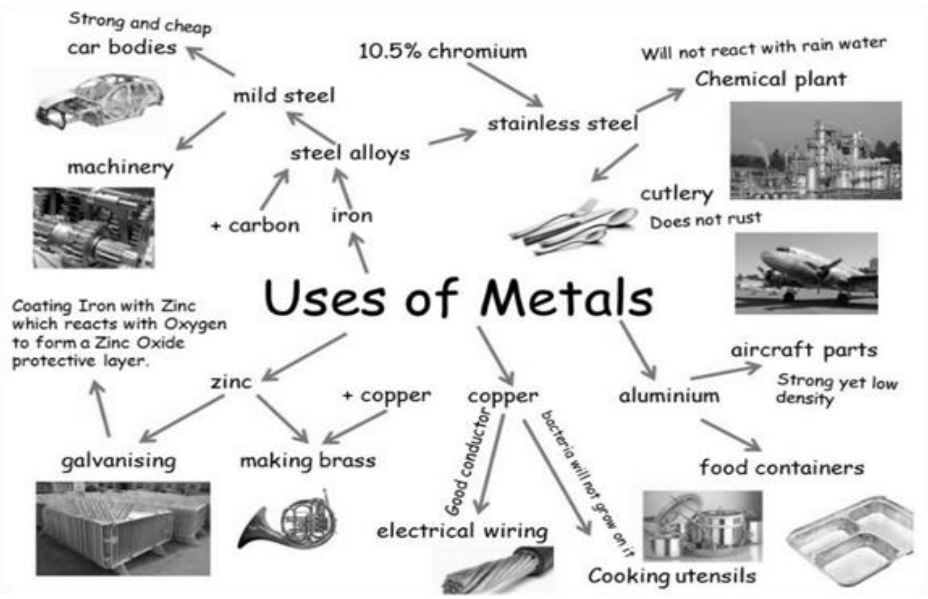
Solids can be divided into two main groups namely metals and non-metals.

METALS	NON-METALS
Iron (Fe)	Wood
Zinc (Zn)	Diamond (made of carbon)
Copper (Cu)	Phosphorus
Aluminium (Al)	Coal (made of carbon)
Gold	Sulphur (S)
Chrome	Carbon (C)
Silver	Nitrogen (N)
Copper (Cu)	Oxygen (O)
Nickel (Ni)	Hydrogen (H)

PROPERTIES OF METALS AND NON-METALS

Properties of Metals	Properties of Non-metals
Most metals are *shiny.*	Non-metals are *dull.*
Most metals are *strong.*	Non-metal solids are *brittle- hard but liable to break easily*
They have high melting and boiling points.	Most of them have lower melting and boiling points.
Metals are good conductors of electricity.	They do not conduct electricity. *Carbon is an exception as it is able to conduct electricity.*
They are good conductors of heat.	They do not conduct heat.
Metals are sonorous-they produce very loud sounds when banged.	They do not produce loud sounds when they are banged.
Metals are ductile- they can be made in to very thin wire without breaking.	
Most metals are solids at room temperature except *mercury.*	
Metals are malleable- *they can be hammered into thin sheets without breaking*.	

USES OF METALS



a) To build things that must carry heavy heights such as bridges and metal beams.
b) They are used for jewellery and the shiny part of motor cars.
c) They are used to make knives for cutting or sharp ends of tools used for digging.
d) Tungsten is used for the thin wire inside an electric light bulb.
e) Copper and aluminium are used for electric cables.

THE USES OF NON- METALS

a) Non-metals like coal and wood are used to produce heat.
b) Phosphorus and sulphur are used to make heads of matches.
c) Diamond is the hardest substance which is used to make drills that can cut through the hardest rock.
d) For the preparation of ammonia, nitric acid and fertilizers, nitrogen is used.
e) For the purification of water, chlorine is used.
f) Hydrogen is very useful as rocket fuel.
g) Carbon can be used to make pencils when it is in the graphite form.
h) Sulphuric acid is prepared using sulphur.

HOUSE HOLD CHEMICALS

Chemicals are substances that can undergo a change or can cause change in a substance. House hold chemicals are substances that we use in the home.

SOLID CHEMICALS

1. Salt- It is made up of small grains and is used to make food taste better.
2. Sugar– it is made up of small grains and it is used for sweetening food and drinks.
3. Baking powder- it produces carbon dioxide when heated and this makes bread or dough rise.
4. Epsom salts- This consists of small crystals .We drinks it to make it to make our stomachs work.

LIQUID CHEMICALS

1. Vinegar– it gives food a sour taste.
2. Oil- it is used for cooking and frying food.
3. Liquid soap- it is used to wash dishes, pots and pans.
4. Liquid cleaners- for example Handy Andy are used to remove dirt and grease.

GAS CHEMICALS

1. Insect sprays are used to kill insects.
2. Bottled gas is used for heating, cooking and lighting.

ENERGY

What is energy?

Energy is the ability to do work. This means when something has energy it can cause something to change. A stone rolling downhill has energy because it is moving.

Forms of energy

a) *Kinetic energy*: This is energy of movement. Any moving object has it.

b) *Heat energy:* All objects contain heat energy. Cold objects have less heat than hot ones.

c) *Potential energy (stored energy):* This is the energy an object has because of its position. A stone that is up a hill has *potential energy*because it can roll down. If the stone starts rolling, it will gain *kinetic energy*. Its potential energy is now converted in to kinetic energy.

d) *Chemical energy:* This is the energy that is released when a chemical reaction takes place (usually when something burns).A piece of wood or coal contains chemical energy.

e) *Electrical energy:* Electrical energy can come from batteries, electric current from a generator or electric power station. Itcan heat water, light a bulb and turn a motor.

f) *Light energy:* A light beam from a very strong light will heat up the surface on which it falls. Light energy is changed toheat energy.

g) *Sound energy:* Sound comes from vibrating objects. For example, the vibrating guitar strings produce music.

SOURCES OF ENERGY

'Source' means where something comes from. Energy comes from many sources.

1. FUELS: A fuel is anything that can produce energy when it burns. Wood, petrol, coal, paraffin and gas are fuels.



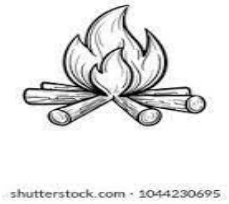
Gas cylinder



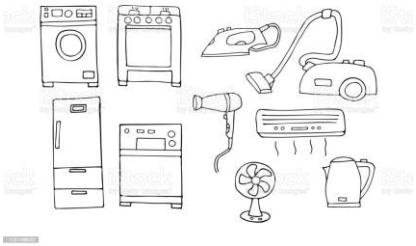
Burning candle



coal



burning wood



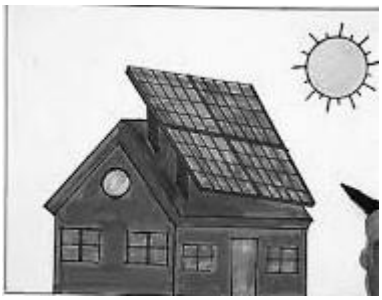
2. ELECTRICITY: It provides energy for lights, electric motors, pumps, machines, electric rains and many other things. Electricity is generated at power stations.

Batteries can also produce electricity for lamps, radios and other appliances.



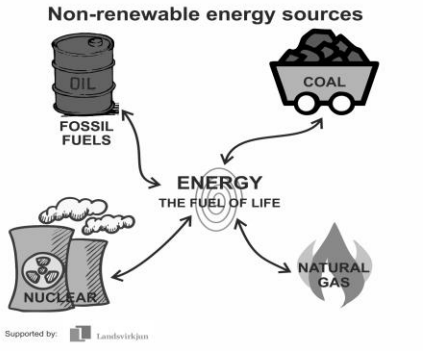
3. WIND: Wind has energy because it can turn a wind mill to pump water.

In other countries there are large wind mills. They turn large generators to produce electricity.



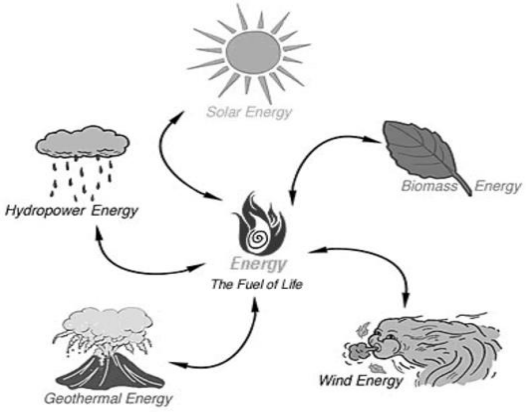
4. THE SUN: All energy on the earth comes from the sun. Plants that died a long time ago used energy from the sun to grow. When they died they got buried deep under the earth and turned to coal, oil and gas. The sun's energy can be converted to electricity through solar cells.

RENEWABLE AND NON-RENEWABLE ENERGY SOURCES



A ***non-renewable energy*** source is one that can be used up. After it is used, it cannot be renewed. Coal, oil and gas are non-renewable sources of energy. Once they are used up they cannot be replaced

A ***renewable source*** of energy is one that is not depleted when used. *Examples are*:
Solar power–energy from the sun.
Wind power–energy from the wind.
Hydroelectric power-electricity generated using water.
Biomass-is plant-based material used as fuel to produce heat or electricity. Examples are wood and wood residues.
Geothermal energy- is *heat within the earth.* It is a renewable energy source because heat is continuously produced inside the earth. People use geothermal heat for bathing, to heat buildings, and to generate electricity.



HOW ORGANISMS OBTAIN THEIR ENERGY
All living things need energy to live.

- THE HUMAN BODY

Our bodies store carbohydrates and fats which combine with oxygen in a chemical reaction called *respiration.* The reaction produces heat and energy for the body to use. Carbon dioxide and heat are produced. The heat keeps our bodies warm and carbon dioxide is exhaled.

- PLANTS

Plants use starch and sugar that is stored as plant food. In a chemical reaction (*respiration*), food is broken down and energy is produced for the plant. Water vapour and carbon dioxide are produced and passed out through the leaves at night.

- GERMS AND BACTERIA

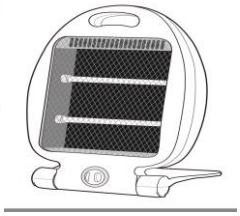
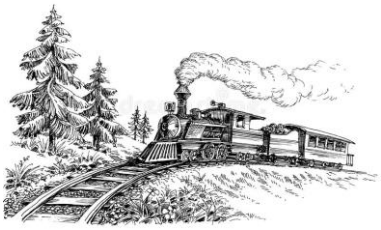
The bacteria that live on something like food or sugar also break up the food in a chemical reaction *(respiration).*

HEAT

What is heat?
Heat is the form of energy that makes the molecules in an object *vibrate*. When something is heated, its molecules vibrate faster and more strongly.

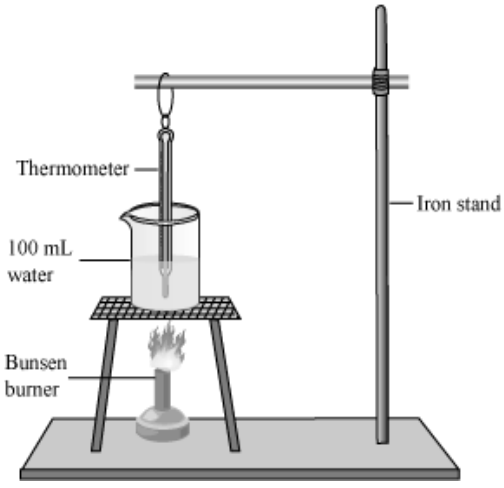
SOURCES OF HEAT
Heat can be generated from:
a) Burning fuel such as wood, coal, oil or gas.
b) Electric current such as in a stove or a heater.
c) The energy of sunlight
d) Friction when materials rub against each other. Fire can be made by rubbing two sticks together.

USES OF HEAT
We use heat to:
a) Cook food
b) Keep our homes and bodies warm
c) Produce steam. The steam can be used to drive machines like locomotives and electrical generators
d) Bending or welding metals. When metals are heated they become softer and can be bent into new shapes.
e) Relieving pain. Putting heat on a sore part of one's body helps lessen the pain.
f) Drying things


EFFECTS OF HEAT
When something is heated, it changes in two main ways:
a) It becomes hotter (its temperature increases)
b) It becomes larger (it expands)

Experiment 1
Temperature increase
When anything gains heat, its temperature increases.
What you will need: a burner, tin/beaker and thermometer
What to do:
1. Pour water into a tin or glass beaker.
2. Place it over a flame.
3. Place a thermometer in the water.
4. Record the temperature every minute.
5. Each time you record, use your finger to feel how hot or cold the water is.
6. What do you notice?

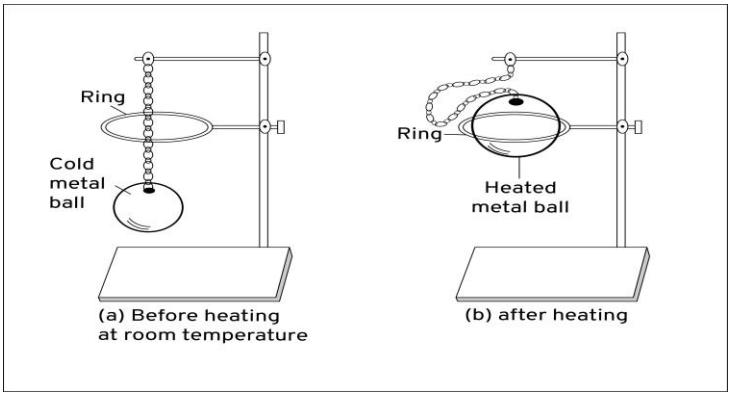


Experiment 2

When something is heated it expands.

What you will need: retort stand, knitting needle and a burner.

1. Find a long knitting needle and clamp its ends in retort stand.
2. Use a burner to heat the whole needle.
3. What do you observe?



LIGHT

What is light?

Light is a form of energy that in straight lines called rays. When light rays enter the eye they produce a sensation called light. Light causes sensation to the *eyes*.

SOURCES OF LIGHT

a) The sun
b) The moon and stars –at night the moon shines reflected sunlight on the earth.
c) Fires and flames
d) Lightning- lightning flashes produces light that lasts for a very short time.
e) Fireflies- these are small insects that fly at night.

KINDS OF LIGHT

There are two kinds of light:

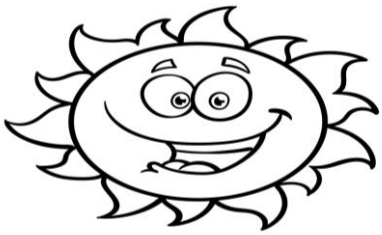
A. Light that is produced naturally (not made by people)
B. Light that is produced artificially (made by people).

NATURAL LIGHT

i) The sun- it produces light and heat.
ii) Lightning –a very large electric spark in the sky that lasts for a very short time.
iii) Fireflies- they produce a flashing light from the back of their bodies.



1 Firefly



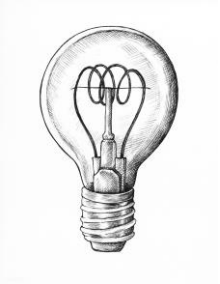


ARTIFICIAL LIGHT

This is light from sources that are made by people.

a) Fire and other flames- anything that burns produces light and heat.

b) Electric light – an electric current heat up the thin wire which glows so brightly that it produces light and heat. The electric current can be from a battery or from the electric current.







LUMINOUS AND NON-LUMINOUS BODIES

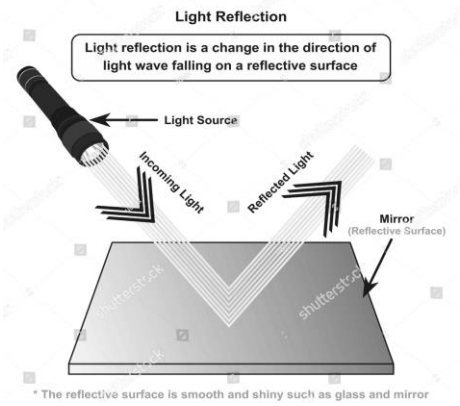
a) LUMINOUS BODIES

These are objects that produce light. The sun, fire, burning lamp and electric light bulb are luminous bodies.

b) NON-LUMINOUS BODIES

These are objects that *do not produce light*. One can see the object because light from another source light shines on it. The moon is a non-luminous body.

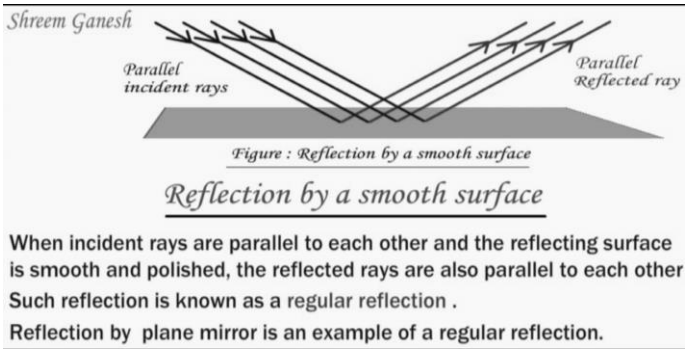
REFLECTION OF LIGHT



When light falls on any surface it seems to bounce off that surface – almost like a ball will bounce off a wall. This is called *reflection*.

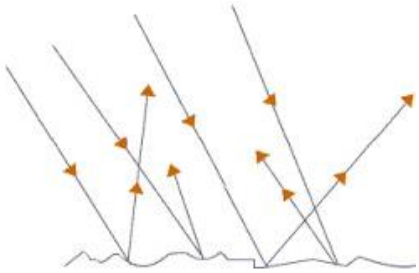
Light is reflected differently by surfaces. *For example*, smooth surfaces will cause regular reflection while rough surfaces will cause reflected light to be scattered.

REGULAR REFLECTION



When light falls on a smooth surface like a mirror, it is reflected back in the same way and the rays stay together.

SCATTERING OF LIGHT



Irregular reflection from rough surface

Light rays fall on a rough surface, they are reflected but they do not stay together. They are scattered. This is called the *scattering of light*.

The scattering of light is also called *irregular reflection*.

SOUND

What is sound?
Sound is a form of energy that travels in waves. It enters the ears and produces a sensation we call sound. We hear sound through our ears.

Some other sources of sound are:
a) People talking.
b) A bell ringing.
c) A car hooting.
d) Thunder roaring.
e) A tin falling off the shelf.

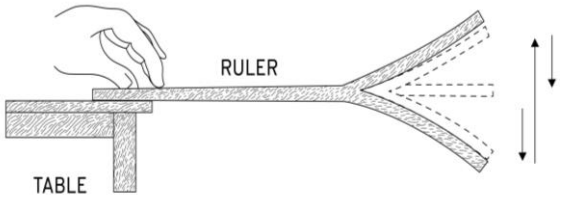


Pleasant and clear sounds are *musical sounds* while unpleasant and harsh sounds are *noise*.

What causes sound?
Sound is caused by vibrating objects.

Experiment
Producing sound
What you will need: a plastic ruler

What to do
1. Hold the ruler over the edge of a table
2. With your other hand, tap the free end of the ruler so that it moves up and down.
3. What do you notice?



A vibrating ruler produces sound

All musical instruments produce vibrations which cause sound.

HOW DOES SOUND MOVE?

When something makes a sound, it travels through the air to our ears. Sound can only travel through a material medium (air, wood, steel, water, glass and others.)

Sound travels better through solids than the air. Sound cannot travel through a vacuum because there is *no air*.

CHANGING SOUND

Experiment

What you will need: a rubber band



Vibrations in a stretched rubber band produces sound

What to do

1. Hold the rubber band in your in your teeth and stretch it hold it in one position.
2. Pluck the rubber band and listen to the sound it makes.
3. Pull the rubber band tighter.

What is the sound like now?

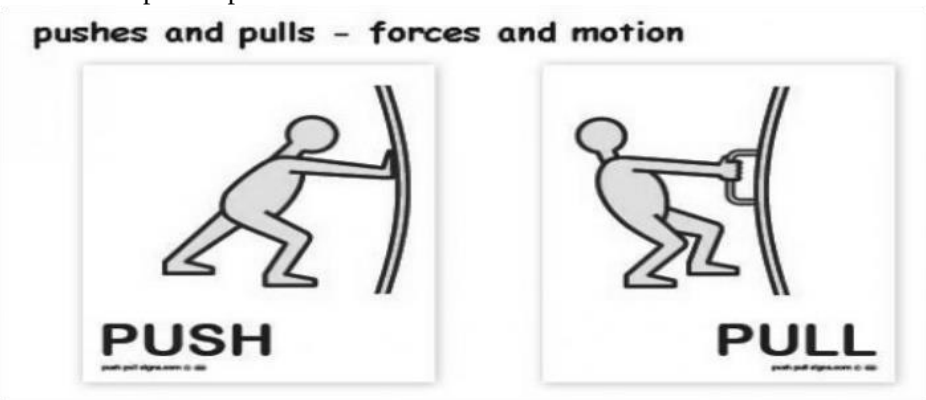
- When the rubber band is pulled tighter it produces a higher note. We say it has a higher pitch.
- Pitch of a musical note is *how high* or *low the note* is.
- We can change the pitch of the sound of a vibrating object by making it vibrate faster or slower.

✈ The faster the vibrations, the higher the pitch and the slower the vibrations, the lower the pitch.

✈ Strong vibrations produce louder sounds while weaker vibrations produce softer sounds.

FORCE

A force is a pull or push.

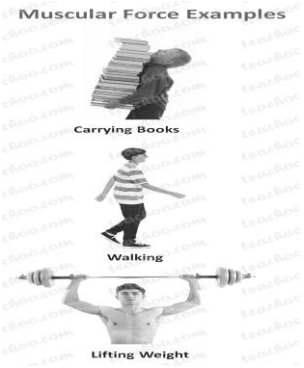


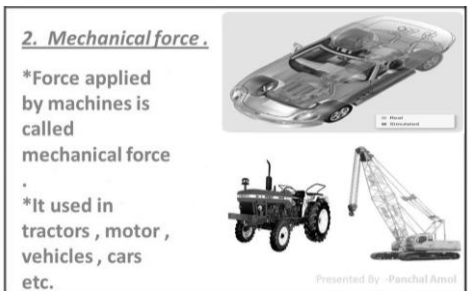
KINDS OF FORCES

There are many different kinds of forces.

a) Muscular force

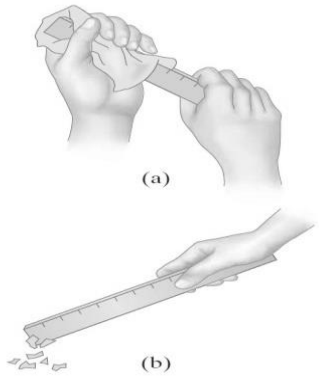
These are forces exerted by people or animals when they use muscles to push or pull.





b) Mechanical Force

These are forces exerted by machines and some kind of motor or engine is used to push or pull.

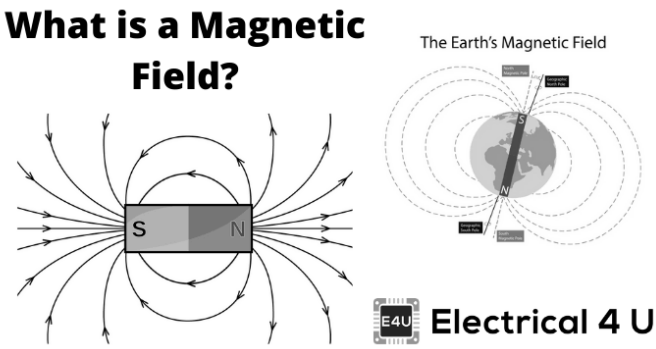


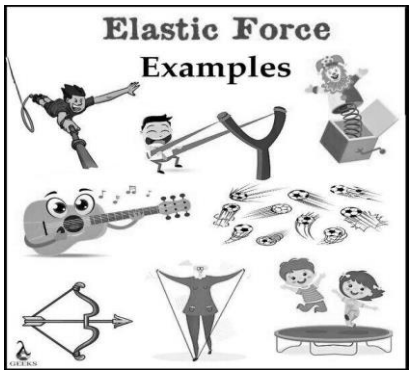
c) Electric force

Rubbing causes certain things to become electrically charged. These charges exert a force we call an electric force.

d) Magnetic force

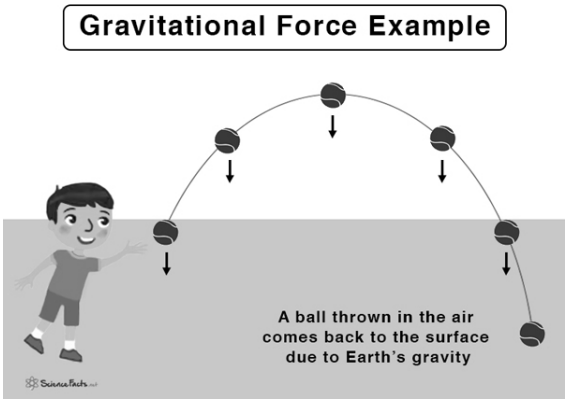
The force between the poles of different magnets is called a magnetic force. The North Pole and South Pole of another magnet attract while the North Pole of the one will repel the North Pole of the other.





e) Elastic force.

When certain things are stretched or squeezed they pull or push back.

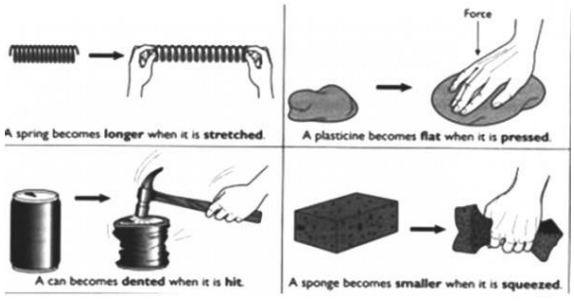


f) Gravitational force

The earth attracts everything and makes it hard to lift things up. This is caused by a force called gravity.

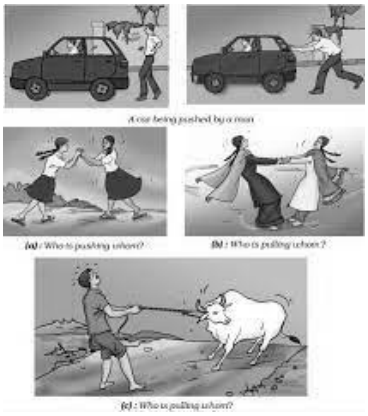
THE EFFECTS OF FORCE

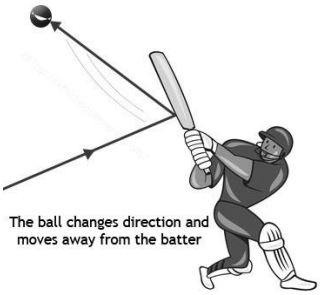
A force can cause changes to an object in the following ways:



- *It can change the shape of an object*

• *It can change the motion of an object*





• *It can change the direction of an object*

MEASURING FORCE



- Force can be measured by something that stretches when it is pulled like a *spring scale* (*force meter)* or rubber band.
- The larger the force, the more the spring or the rubber band stretches.